

GEOMIND-R

Generative inverse design of geopolymers for the immobilisation of radioactive Cs-137 and Sr-90

First formal progress report to the supervising chemistry professor — pre-seminar checkpoint covering the chemical foundations, results achieved, the data-integrity audit, current obstacles, and the plan forward.

Principal investigator	Mulham Fetna · ORCID 0009-0006-4432-798X
Supervisor	Dr. Abdulrazzaq Hammal · ORCID 0000-0003-1828-1376
Institution	[Institution / Department]
Seminar date	21 July 2026
Report status	Checkpoint 1 · literature-scale meta-analysis phase

Executive summary

Where the project stands

The project set out to replicate the *GEOMIND* generative model for geopolymer design and to extend it with a fourth design dimension — the capacity of a geopolymer to capture and retain radioactive caesium and strontium. Over this first phase the chemical groundwork has been laid, a compiled dataset has been built and rigorously audited, and one genuinely novel chemical result has emerged. The project is *not* yet able to train the generative extension, for reasons that are now understood to be structural rather than incidental — and that diagnosis is itself one of the phase's most useful outcomes.

126

Pool A rows (adsorption) · 11 sources

40

Pool B rows (immobilisation) · 4 sources

26

audited findings (19 resolved)

122

automated integrity tests passing

THE THREE THINGS A READER SHOULD TAKE AWAY

- A real chemical finding.** For the uptake of cations such as Cs^+ and Sr^{2+} , *framework aluminium* — the tetrahedral Al^{IV} that creates the charge-balancing exchange sites — predicts capacity strongly (correlation up to **+0.95**), whereas *surface area* (BET), the quantity most often reported, does not (pooled **+0.08**). The two even move in opposite directions for organic dyes, confirming a different binding mechanism. This reframes what a "good adsorbent geopolymer" is.
- A rigorous data-integrity audit.** The compiled starting dataset was found, on source-by-source checking, to contain roughly **one-third** unreliable rows — including a europium measurement mislabelled as strontium, a fabricated feasibility domain, transplanted columns, and a published adsorption capacity a thousandfold larger than the same paper's measured uptake. Every one was traced, evidenced, and quarantined. The audit trail is a contribution in its own right.
- An honest obstacle.** The single descriptor that predicts uptake — framework Al — is almost never reported *together with* a Cs/Sr capacity in the same study. This is a structural feature of how two separate research communities publish, not a gap our searching can close. It is the reason the generative extension cannot yet be trained, and it defines exactly which papers must be acquired to proceed.

Contents

Structure of this report

1	Aim and scientific framing	8	Plan and next steps
2	Chemical foundations	A	Full finding register (all findings)
3	Achievements — chemical findings	B	Pool A — every adsorption row
4	Achievements — the data-integrity audit	C	Pool B — every immobilisation row
5	Current data structure	D	Decisions register
6	What we could not achieve, and why	E	Glossary of chemical terms
7	Technical / machine-learning progress		

Section 1

Aim and scientific framing

Geopolymers are *alkali-activated aluminosilicates*: an aluminosilicate precursor (metakaolin, fly ash, or blast-furnace slag) is dissolved in a strongly alkaline activator (NaOH or KOH with sodium silicate) and re-condenses into a rigid three-dimensional network of corner-sharing SiO_4 and AlO_4 tetrahedra. Because aluminium sits in *tetrahedral* coordination (Al^{IV}), each AlO_4 unit carries a net negative framework charge, balanced by a mobile alkali cation. Those charge-balancing sites are the ion-exchange sites that make a geopolymer able to capture cationic radionuclides such as Cs^+ (**caesium-137**) and Sr^{2+} (**strontium-90**) — two of the most mobile and long-lived fission products in nuclear waste.

THE REFERENCE WORK BEING REPLICATED AND EXTENDED

The starting point is *GEOMIND* (RSC *Digital Discovery*, 2026, DOI 10.1039/d5dd00383k), a hybrid generative model that performs *inverse design*: given a set of target properties, it generates a candidate geopolymer *formulation*. It couples two variational autoencoders — a **Simulator** (formulation → properties) and a **Formulator** (properties → formulation) — with a **feasibility block** that keeps generated formulations chemically viable.

THE PROJECT IN ONE SENTENCE

Reproduce GEOMIND, then add a **fourth design dimension** — the capacity of a geopolymer to capture and retain Cs⁺ and Sr²⁺ — so that formulations can be generated specifically for radionuclide immobilisation.

A DISTINCTION THAT RESHAPED THE PROJECT: TWO DIFFERENT CHEMISTRIES

Early work revealed that "capturing" a radionuclide is not one property but two, with different chemistry and opposite figures of merit:

Adsorption	The geopolymer is placed in a <i>solution</i> and removes the cation from it. The target is a capacity (mg g ⁻¹ , or a distribution coefficient <i>K_d</i>). Higher is better. — <i>Pool A</i>
Immobilisation	The cation is <i>built into</i> the solidified waste form and must not escape. The target is a leach resistance (leachability index, effective diffusivity). Higher resistance is better. — <i>Pool B</i>

These are different applications answering different questions, and — as Section 6 explains — the chemical descriptor that predicts one is reported almost exclusively in studies of the other. On the supervisor’s decision the two are kept as **two parallel datasets, never merged** (project decision D13).

HONEST SCOPE OF THE DATASET

The compiled dataset is *literature-derived*, not newly measured in the laboratory. Consequently every claim is framed as a **literature-scale meta-analysis and external validation**, never as independent experimental replication. This framing is stated up front because it governs how strongly any result may be worded.

Section 2
Chemical foundations

This section sets out the chemistry the rest of the report depends on. It is written for a chemist and is deliberately the most detailed part of the document.

2.1 The geopolymer network and its descriptors

The reactive network is described by a small set of chemical descriptors, each of which the project treats as a candidate predictor of radionuclide uptake.

THE SI/AL RATIO

The molar **Si/Al** ratio is the primary compositional lever. A lower Si/Al means more AlO₄ tetrahedra, hence more framework charge and more cation-exchange sites — so, all else equal, lower Si/Al should mean higher cation capacity. It is the most-reported descriptor, and precisely because it is a bulk ratio it is also the easiest to get wrong (see the audit, Section 4).

SILICON SPECIATION: THE Q^N(MAL) NOTATION

Solid-state ²⁹Si magic-angle-spinning NMR resolves the local environment of each silicon atom. In the notation **Qⁿ(mAl)**, *n* is the number of bridging oxygens on the central Si (0–4, i.e. how condensed the site is) and *m* is how many of those bridges connect to an *aluminium* rather than another silicon. A fully condensed, aluminium-rich site is Q⁴(4Al); a fully condensed, aluminium-free site is Q⁴(0Al). Sites with non-bridging oxygens (Q³, Q²...) indicate a less-polymerised, more disrupted network.

FROM SPECIATION TO A FRAMEWORK-ALUMINIUM DESCRIPTOR (ARI)

Weighting each silicon environment by its aluminium count and summing gives the mean number of Al neighbours per Si — the quantity the project calls the **Al-richness index**:

ARI = Σ_m (m · I_m) / 100 (I_m = % of Si in a site with *m* Al neighbours)

ARI is proportional to the density of tetrahedral-Al charge-balancing sites — it is a direct, NMR-based measure of the framework aluminium that ion exchange actually uses, without needing ²⁷Al NMR. It is the central descriptor of this project.

BET SURFACE AREA — THE DESCRIPTOR THE FIELD USUALLY REPORTS

The **BET** specific surface area (m² g⁻¹, from N₂ physisorption) is the quantity most adsorption papers foreground. It measures physical surface available for physisorption. A central result of this project (Section 3) is that for *cations* it is the *wrong* descriptor.

2.2 The immobilisation chemistry and the leachability index

In a waste form the radionuclide is incorporated during synthesis and its escape is measured by a semi-dynamic leach test. The transport is summarised by an effective diffusivity *D_e* (cm² s⁻¹) and, in the ANSI/ANS 16.1 standard, by the dimensionless **leachability index**:

L (or LI) = -log₁₀(*D_e*) – higher LI = slower leaching = **better** immobilisation;
LI ≥ 6 is the regulatory acceptance floor.

Because LI is logarithmic, a mean LI over several leaching intervals is a *geometric* mean of *D_e* and cannot be inverted to a single diffusivity — a subtlety the data schema enforces, and which was independently confirmed against three separate published tables. A second, more dangerous subtlety: the phrase "diffusion coefficient of Cs" is used in the literature for *two opposite experiments* — the cation leaching *out* of a doped waste form (10⁻¹⁶...10⁻¹⁹ cm² s⁻¹) versus diffusing *into* an immersed specimen from solution (~10⁻⁸ cm² s⁻¹). The two differ by **twelve orders of magnitude**; the dataset keeps them as distinct quantities that may never be pooled.

2.3 The adsorption isotherm and a saturation screen

Adsorption capacities are usually reported as a Langmuir maximum, $Q_{\max}$, fitted to an isotherm. But a fitted $Q_{\max}$ is only a real capacity if the experiment actually approached saturation. At low surface coverage the Langmuir form is nearly linear, so it fits the data beautifully (high R^2) while the plateau it implies is essentially unconstrained — an authoritative-looking number that means very little.

The project therefore screens every fitted capacity by its fractional surface coverage at the highest concentration used:

$$\theta = b \cdot C_0 / (1 + b \cdot C_0) = 1 - R_L \quad (R_L = \text{Langmuir separation factor, often already reported})$$

$\theta \geq 0.8$ means the plateau was genuinely approached; $\theta < 0.5$ means the "capacity" is an extrapolation. The identity $\theta = 1 - R_L$ means the screen usually needs no extra data at all. This check — to our knowledge not routinely applied when capacities are compiled across studies — is one of the project’s methodological contributions (Section 3).

Section 3

Achievements — chemical findings

3.1 The central result: framework aluminium predicts cation uptake; surface area does not

Across every dataset in which the two can be separated, the density of framework aluminium — measured as Al^{IV} content or as ARI — tracks cation uptake strongly and positively, while BET surface area does not. Pooled across all cation datasets the BET correlation collapses to essentially zero.

Evidence set	n	corr(BET, uptake)	corr(framework Al, uptake)
Varon 2025 (Sr, K _d)	7	+0.19	+0.95
Oulu Ca-free series (NH ₄ ⁺ , ARI)	4	−0.98	+0.93
Xiang foams (Cs)	5	−0.90	—
Lin 2026 (cation)	4	−0.90	—
Oulu NH ₄ ⁺ (all)	21	−0.47	—
Pooled cations	—	+0.08	—
Organic dyes (contrast class)	42	+0.66	—

WHY THIS IS CHEMICALLY SENSIBLE

Cation uptake here is **ion exchange at charge-balancing AlO₄ sites**, not physisorption on a surface. More framework Al = more exchange sites = more capacity, regardless of how much physical surface the material presents. The contrast class proves the mechanism: for **organic dyes**, which *do* bind by surface physisorption, the BET correlation flips to **+0.66**. The descriptor that matters depends on the binding mechanism — so cation and dye data must never be pooled into one target.

3.2 A structural precondition on the descriptor (finding F19)

The framework-aluminium descriptor is not universal: it only means something where a condensed aluminosilicate framework exists. When appreciable **calcium** is present, Ca acts as a *network modifier*, disrupting the silicate linkages and driving the silicon speciation toward low-connectivity Q⁰–Q² environments — in the calcium-bearing samples analysed, the fully-condensed Q⁴ population fell essentially to zero. With no framework to measure, *no* compositional descriptor predicts uptake (ARI, Si/Al, Ca/Al and BET all $|r| \leq 0.46$), and capacity is nearly flat at $7.3 \pm 1.3 \text{ mg g}^{-1}$. Pooling calcium-rich and calcium-free sorbents together produced a misleadingly mediocre correlation that concealed two entirely different regimes. The lesson — *never pool across structural classes* — generalises the dye/cation lesson one level deeper.

3.3 The isotherm saturation screen (findings F14–F17)

Applying the $\theta = 1 - R_L$ screen of Section 2.3 to all 35 fitted Langmuir capacities in the pool:

31

sound ($\theta \geq 0.8$) — measured, not extrapolated

1

borderline ($\theta = 0.78$)

3

extrapolation artefacts, excluded

The trigger case: a chabazite fit reported $Q_{\max} = 1249.5 \text{ mg g}^{-1}$ at $R^2 = 0.99989$, while the same paper’s measured equilibrium uptake was **1.184 mg g^{−1}** — a capacity extrapolated a thousandfold beyond any data point, flagged by nothing in the reported statistics. Two of the excluded fits additionally showed a negative intercept in their linearised form, implying an unphysical negative affinity constant. When a sixth source was later recovered (Section 4), the screen immediately caught a further artefact — the *highest* reported capacity of that source’s five, unsaturated at $\theta = 0.50$.

3.4 Immobilisation: strontium is held far better than caesium

In the immobilisation pool a single result is strikingly robust: in the *same* binder, strontium is retained far better than caesium. In one metakaolin K–A–S–H gel the effective diffusivities differ by roughly **six orders of magnitude** ($D_{e,Sr} \approx 8 \times 10^{-20}$ vs $D_{e,Cs} \approx 6 \times 10^{-14} \text{ cm}^2 \text{ s}^{-1}$), and Sr’s leachability index exceeds Cs’s in *every* one of the fourteen specimens measured. Chemically: Sr²⁺, being divalent and readily precipitated as SrCO₃, is chemically bound in the gel, whereas the large, singly- charged, highly soluble Cs⁺ is held only weakly. Every alkali-activated matrix in the pool clears the ANSI/ANS 16.1 acceptance floor ($LI \geq 6$) for both nuclides.

3.5 A methodological result: a rejected NMR deconvolution that was rejected for the right reason

An attempt to derive the Q⁴(mAl) speciation of a set of ²⁹Si spectra by fitting five Gaussian peaks reached an excellent fit ($R^2 \approx 0.998$) yet was *rejected*, because it forced the Q⁴(3Al) population to exactly zero — which is chemically impossible for the composition. The rejection was later vindicated when the authors’ own deconvolution was found already published: they use **ten** silicon environments including Q³/Q² species, which a five-peak, Q⁴-only model structurally cannot represent. A chemical sanity check caught a structurally wrong model that a near-perfect fit statistic could not — a cautionary result worth reporting in its own right.

Section 4 Achievements — the data-integrity audit

The largest single body of work in this phase was verifying the compiled data against its primary sources. It is presented candidly because it is, scientifically, the most defensible contribution made so far: **roughly one-third of the compiled rows did not survive source-by-source audit**. A representative sample of what was caught:

What was found	Detail
Europium mislabelled as strontium	A Eu measurement carried into the pool as an Sr capacity — different element, different chemistry.
Fabricated feasibility domain	A ‘feasibility range’ table attributed to GEOMIND was in fact copied from this project’s own retired heuristic tool; applied to the paper’s own samples it rejected 7 of 9 feasible and accepted the 1 infeasible.
Transplanted columns	Si/Al and density values from the paper’s 10 published samples had been broadcast across an unrelated concrete dataset; both columns dropped.
A thousandfold-extrapolated capacity	The chabazite $Q_{\max}$ of 1249.5 vs 1.184 mg g ^{−1} measured (§3.3).
Misattributed EXAFS values	Two strontium coordination numbers credited to one author were in fact another study’s, verbatim.
A published typo, caught by arithmetic	A tabulated leachability index (15.5) was inconsistent with its own diffusivity; the paper’s own column average proved the intended value was 15.1.
Silent half-table loss	A provided workbook had dropped both caesium rows of a source’s leach table — half the data — undetectably without reading the source.

THE STANDING RULE THIS PRODUCED

No number enters either dataset until its value has been read *in context* in the primary source PDF. A number existing somewhere in a paper is not evidence of what it means. Every quarantine and correction is enforced in code and covered by an automated test, so a future edit cannot silently reintroduce a rejected value.

THE AUDIT ALSO TURNED UP VALUE, NOT ONLY ERRORS

Reading sources in full — rather than trusting a keyword-based triage — recovered several papers that had been wrongly set aside. One rejected "diffusivity study" in fact published caesium adsorption capacities for five different activators (a genuine gap-filler for Pool A); another "review" turned out to index exactly the paired-descriptor sources the project most needs; and two of the most descriptor-rich papers held were initially invisible because their filenames were not in Latin script. Four times this phase, the most valuable next step proved to be reading a document already in hand rather than acquiring a new one.

The complete, evidenced register of all 26 findings (19 resolved) is given in Appendix A.

Section 5 Current data structure

All data lives in a single governed system, not in scattered files. Three layers keep it honest.

5.1 A single source of truth (the registry)

Every data asset, primary source, finding and decision is recorded in one machine-checked registry (17 assets, 17 sources, 26 findings, 16 decisions). A validator rejects any dangling reference, unknown status, or finding without an action, and regenerates the human-readable documentation from it. The registry — not anyone’s memory — is the project’s ground truth.

5.2 A strict labelling system

Every mined record is tagged on three independent axes, so a record’s trustworthiness is explicit and a model can be restricted to only defensible rows:

Veracity	verified-true · probable · unverified · redundant · false
Utility	core · supporting · context · discard
Granularity	observation · raw-signal · reference

Only rows that are (verified-true or probable) **and** (core or supporting) **and** an observation are considered modellable.

5.3 Two parallel datasets (decision D13)

The two chemistries of Section 1 are held as two schemas that share no target column, so they can never be concatenated by accident.

POOL A — ADSORPTION (126 ROWS, 11 SOURCES)

Target: adsorption capacity (mg g^{−1}) or distribution coefficient. The schema separates five distinct capacity *types* (a fitted Langmuir maximum and a single-point uptake are not the same number) and four adsorbate classes, so incompatible quantities cannot be silently averaged.

Source	rows	Adsorbate class	rows
oulu2026	63	radionuclide	60
literature_compilation	21	dye	42

Source	rows	Adsorbate class	rows
varon2025	7	analogue	21
tarnovsky2024	6	other	3
lin2026	6		
katada2024	6		
kurumisawa2021	5		
xiang2021	5		
baek2018	3		
tian2019_espr	2		
zhang2021	2		

POOL B — IMMOBILISATION (40 ROWS, 4 SOURCES)

Target: leach resistance. The schema encodes four traps peculiar to this chemistry: the figure of merit is *inverted* (a high leach rate is bad); doped loading is a *control* variable, never a target; a below-detection result is *information* (the best performers), kept as left-censored data rather than discarded; and the two ‘diffusion coefficients’ that differ by twelve orders of magnitude are kept as separate quantities.

Source	rows
jang2016	14
kurumisawa2021	10
nevin2026	10
kim2026	6

Every individual row of both pools is listed in Appendices B and C.

Section 6

What we could not achieve, and why we are stuck

The generative extension — the project’s ultimate goal — cannot yet be trained. The reasons are specific and, importantly, now diagnosed rather than merely encountered.

6.1 The predictive descriptor is structurally scarce (finding F20)

Framework aluminium is the one descriptor that predicts cation uptake (Section 3.1) — and it is reported almost nowhere alongside a Cs/Sr capacity. A content survey of the whole corpus made the reason plain:

TWO COMMUNITIES THAT DO NOT OVERLAP

Every adsorption study in the corpus reports zero solid-state NMR, while four of the six NMR-rich papers are immobilisation (waste-form) studies. Water-treatment groups measure isotherms and surface area and never run NMR; nuclear-waste groups run full ²⁹Si/²⁷Al NMR and measure leach rates and never run isotherms. The descriptor and the target sit on opposite sides of a disciplinary divide.

The practical consequence is decisive: **more literature searching cannot fix this** — the co-occurrence is rare by the structure of the field, not by any narrowness of our search. Framework-Al data currently exists paired with a capacity in only a handful of rows, from a single source. Both datasets end up descriptor-starved, for mirror-image reasons: Pool A has composition but no NMR; Pool B has NMR but almost no compositional variation.

6.2 The faithful replication is blocked on confidential data

GEOMIND was trained on 112 samples from one laboratory under one protocol; only 10 are public. A *faithful* replication (milestone M3a) is impossible without the full set, which is confidential. A formal data-request letter has been drafted and awaits sending. Our own compiled data cannot substitute: it is fly-ash/slag concrete, whereas GEOMIND is metakaolin paste — a different material system with no viscosity target and only ~92 distinct mix designs, so that strand has been honestly *re-scoped* from ‘replication’ to a ‘cross-system transfer study’.

6.3 Inverse design is not reachable on the present data

There are only 60 radionuclide (Cs/Sr) rows in Pool A, spread across many laboratories and protocols. Training a generative inverse-design model on so few rows from so many sources would learn inter-laboratory protocol differences and report them as chemistry — precisely the failure the audit exists to prevent, one level up. The defensible near-term output is a *descriptor-and-methodology* study, not a generative model.

THE OBSTACLE STATED PLAINLY

We are not stuck for lack of effort or data volume. We are stuck because the one chemical descriptor that works is, by the structure of the published literature, almost never measured together with the property we want to predict. That diagnosis tells us *exactly* which few papers would unblock the work (Section 8).

Section 7

Technical / machine-learning progress (high-level)

For completeness, the engineering status at a glance. The chemistry above is where the scientific progress lives; the technical layer is deliberately kept faithful and modest.

Milestone	State	Note
M1 — paper analysis	done	Reproduction specification written; the authors' code located (unlicensed → clean-room only).
M2 — data cleaning & audit	done	~32% of compiled rows failed source audit (Section 4).
M3a — chemistry layer	verified	Si/Al, Si/M _{sol} and Solid/Liquid molar ratios reproduce the paper’s own samples to within 1–3%.

Milestone	State	Note
M3a — generative VAEs	not started	Blocked on training decisions and, more fundamentally, on data (Section 6).
M3b — cross-system transfer	re-scoped	Not a GEOMIND validation — a different material system, stated as such.
M4 — adsorption dimension	in progress	The chemical question is answered; the descriptor is scarce.
M5 — radionuclide design	blocked	Needs M4.
M6 — corpus expansion	continuous	Automated scraping exhausted; targeted acquisition now.

The one piece of the model already built and *verified* is the chemistry layer: the molar-ratio calculations that any generated formulation must satisfy reproduce the authors’ published samples to 1–3%. This is the foundation the generative blocks would sit on. Everything is version-controlled with a full branch/issue workflow, and the whole data pipeline is covered by **122 automated tests** that encode each chemical rule (unit conversions, the saturation screen, the pooling prohibitions, the leachability-index mathematics) so that no future change can quietly violate them.

Section 8

Plan and next steps

8.1 Immediate: acquire the few papers that unblock the descriptor

Section 6 turns an open-ended obstacle into a short shopping list. These specific sources are the ones identified as carrying framework-Al data *together with* a Cs/Sr retention or capacity measure — the combination the field almost never publishes.

Priority	Source to acquire	Why it matters	Unblocks
1	Geddes <i>et al.</i> 2025, <code>J. Hazard. Mater.</code> 488	Title promises Sr incorporation mechanism (NMR) and binding capacity in one paper — the rare bridge of Section 6.	M4, M5
1	Komljenović <i>et al.</i> 2020, <code>J. Hazard. Mater.</code> 388	Ties ²⁹ Si speciation to strength and Cs immobilisation — a second candidate bridge.	M4, M5
2	Qian <i>et al.</i> 2001, <code>J. Nucl. Mater.</code> 299	Cs and Sr distribution ratios (K _d) — primary data for Pool A.	M4
2	<i>Next Sustainability</i> 2025	The only source with a 1–5 M activator-concentration axis alongside Cs, Sr and NMR.	M6
3	GEOMIND 112-sample dataset (confidential request)	Larger than every scraping avenue combined; the only route to a faithful replication.	M3a

A consolidated worklist, with where each file should be placed, is maintained at `docs/acquisition-worklist.md`.

8.2 Near term: the defensible scientific output

Independent of acquisition, the present data already supports a **descriptor-and- methodology paper**: that for cation uptake framework aluminium predicts and surface area does not, with the structural precondition (Section 3.2); and that a simple saturation screen ($\theta = 1 - R_L$) exposes published capacities that are extrapolation artefacts (Section 3.3). Both are novel, both are defensible now, and neither requires the confidential dataset.

8.3 Medium term: the two-pool path to the design goal

With the priority-1 papers in hand, the framework-Al descriptor can be paired with a retention target in enough samples to test whether it also predicts *immobilisation* (it should, by the same charge-balancing chemistry). That is the bridge from the descriptor result to the project’s stated aim of designing formulations for Cs-137/Sr-90. The generative model (M3a/M5) remains gated on either the confidential GEOMIND set or a substantially enlarged paired dataset — and the report is deliberately honest that this is not within reach on current data.

8.4 What we are asking of this checkpoint

- Agreement that the **descriptor-and-methodology result** is the right near-term scientific target.
- Guidance on **acquiring the priority-1 papers** (institutional access to `J. Hazard. Mater.`), and on whether the department can support the confidential GEOMIND data request.
- A steer on whether a small **own-laboratory NMR measurement** on a few sorbents of known capacity is feasible — the only route that would generate genuinely new paired data rather than re-using the literature.

Appendix A

Full finding register

All 26 audited findings, most-recent last, each with the evidence that established it and the action taken. Severity is the data-integrity impact.

critical	F1 — Si_Al and Density transplanted from the GEOMIND paper [resolved]
EVIDENCE	
9 of 11 unique Si_Al values match the paper's 10 samples exactly to 3 dp; all 10 published density values appear; 91 of 92 mixes carry a single broadcast value; neither column exists in either upstream source file.	
ACTION	
Both columns dropped in clean.py. Must never be reinstated as features.	
critical	F2 — Dataset far smaller than 867 rows implies [resolved]
EVIDENCE	
Only 92 distinct mix designs; Metakaolin constant zero; features have 2-6 unique values.	
ACTION	
Grouped CV by mix design is mandatory. Metakaolin dropped.	

high F3 — Adsorption data has extensive internal duplication [resolved]

EVIDENCE

24 Qmax_Cs rows contain only 16 distinct values; Qmax_Sr has 7 rows.

ACTION

Repeats flagged (not deleted) via cs/sr_value_repeated.

medium F4 — Redundancy and version disagreement [partially resolved]

EVIDENCE

CSV/XLSX byte-identical; 18 duplicate rows; v2 disagrees in 22 cells.

ACTION

CSV declared canonical; duplicates removed; v2 unresolved.

low F5 — Metadata.csv is cp1256, not UTF-8 [resolved]

ACTION

Handled in the ingestion layer.

critical F6 — Feasibility_Ranges is mislabelled and anti-correlated with truth [resolved with open gap]

EVIDENCE

Rejects 7 of 9 feasible published samples, accepts the 1 infeasible. Its constants are verbatim from the retired tool at tag v1.0.0.

ACTION

Sheet not read. Provisional envelope used instead, flagged provisional=True.

critical F7 — Compiled data failed source audit - 10 of 31 rows [resolved]

EVIDENCE

(a) 3 rows labelled Sr are Eu-152+154. (b) 4 Niu 2022 rows claim Langmuir Qmax from a paper tabulating none. (c) Na-MK misattributed from another paper. (d) K-MK is a verbatim copy of K-MKBFS - a sorbent that does not exist.

ACTION

Quarantined/relabelled/re-typed as named constants in code.

medium F8 — Unmined master sheets add nothing [resolved]

EVIDENCE

Adsorption_Isotherms and Cs_Isotherms duplicate Functional_Data with the same fabricated rows; only 3 new values, which are cobalt.

ACTION

Marked duplicate. Not ingested.

high F9 — Surface area does not predict CATION uptake [open]

EVIDENCE

Cation-only, per source - lin2026 -0.90 (n=4), xiang2021 -0.90 (n=5), oulu2026 -0.47 (n=21), tarnovsky2024 +0.22 (n=6, inconclusive). Pooled cations +0.08. Varon: BET +0.19 vs Al-IV +0.95.

ACTION

Al-IV is the candidate primary feature. NOTE - an earlier claim of "six datasets confirm a negative BET correlation" was an OVERSTATEMENT, corrected here.

high F10 — Pooling adsorbates flips the sign of the BET relationship [resolved]

EVIDENCE

cations corr(BET,capacity) = +0.08; dyes = +0.66.

ACTION

Never pool adsorbate classes. Dyes use surface area; cations use charge sites. pooling_warning() enforces the check.

high F11 — Sr_Immobilization_Mechanisms is untraceable and largely duplicated [resolved]

EVIDENCE

Audit 2026-07-20 - the ONLY master sheet with no Source column at all, plus 66.7% internal duplication among its numeric cells. Same signature as the fabricated Na-MK/K-MK rows in F7.

ACTION

Relabelled FALSE / DISCARD in the warehouse. Not usable.

medium F12 — The v2 "conflict" was precision, not disagreement [resolved]

EVIDENCE

All 22 differing cells are Comp_Strength in rows 843-866 and differ only by rounding (34.36 vs 34.35777). N data.xls genuinely reports up to 6 dp, and 18 of the 24 v2 values appear verbatim upstream. So v2 preserved source precision and Physical_Data rounded to 2 dp.

ACTION

Q1 closed. v2 status suspect -> verified. Physical_Data remains canonical; the difference (<=0.005 MPa) is immaterial for modelling.

medium F13 — Several context sheets cite papers we do not hold [open]

EVIDENCE

Audit 2026-07-20 - Kim_2026, Yildirim_2024 and Duque_Redondo_2023 are cited by Surface_Complexation_Sr, MD_Simulation_Sr_Cs, Zeolite_Comparison_Sr_Cs, Sr_Leaching_Data, EXAFS and XRD sheets. None are in papers/references. Surface_Complexation cites "Eq. (10)-(12)" - equations, not sources, so those rows are model output rather than measurement. FTIR is 88.9% duplicated, XRD 72.7%.

ACTION

All remain UNVERIFIED and excluded from modelling. Acquiring the three papers is the only way to promote them; recorded as Q6.

high F14 — A high R2 Langmuir fit can leave Qmax essentially unconstrained [open]

EVIDENCE

Baek 2018 chabazite: Q0L = 1249.5 mg/g with K_L = 7.9e-05 L/mg and R2 = 0.99989, while the paper's own kinetics report equilibrium uptake of just 1.184 mg/g. K_L is two orders of magnitude below the other two zeolites in the same table. At low coverage the Langmuir isotherm is near-linear, so it fits almost perfectly while the plateau is extrapolated ~1000x beyond any measured point.

ACTION

Ingested but flagged - it exceeds the schema bound of 1000 mg/g and validate() reports it. Must be excluded from any fit unless the extrapolation is defended. Generalises: R2 is not evidence that a fitted Qmax is meaningful. Screen future Langmuir rows on K_L and on the measured concentration range, not R2 alone.

critical F15 — Fitted capacities cannot be validated - C0 was never captured [resolved with open gap]

EVIDENCE

Applying the F14 screen to all 35 fitted-isotherm rows: 32 of the 33 rows with a usable K_L have NO recorded initial concentration, so Langmuir saturation ($\theta = K \cdot C_0 / (1 + K \cdot C_0)$) cannot be computed. The screen runs on exactly one row (GF600, $\theta = 0.93$). Chabazite was caught only because its paper stated the equilibrium uptake in prose.

ACTION

initial_conc_mg_L made required for fitted-isotherm rows going forward; existing rows flagged c0_unknown. PROGRESS 2026-07-20: C0 recovered for Baek (2658 mg/L) and El-Naggar (1000 mg/L), taking screenable rows from 1 to 10 - of which 8 are sound, 1 borderline (stilbite, $\theta = 0.775$) and 1 a confirmed artifact (chabazite, $\theta = 0.174$). 25 rows still lack C0. COMPLETE 2026-07-20: C0 recovered for 7 sources; 29 of 35 rows now screenable. 27 sound, 1 borderline (stilbite 0.775), 1 artifact (chabazite 0.174). Six rows remain unscreenable because their methods state no isotherm range. Lesson recorded THREE times: K_L alone is not a screen - saturation is $K \cdot C_0 / (1 + K \cdot C_0)$, so a small K is compensated by a large C0. The same instinct would have discarded 19 sound rows.

medium F16 — $\theta = 1 - R_L$, so the saturation screen needs no C0 [resolved]

EVIDENCE

The Langmuir separation factor $R_L = 1 / (1 + b \cdot C_0)$ is reported by many papers as a favourability check. Our saturation measure $\theta = b \cdot C_0 / (1 + b \cdot C_0)$ is exactly $1 - R_L$. Verified against the independent C0 route on six El-Naggar rows: the two derivations agree to three decimals. Resolved Tian ESPR ($\theta = 0.944$ Cs, 0.994 Sr) with back-calculated C0 5.25 and 9.77 mmol/L, matching the Fig. 5 axes.

ACTION

Use R_L as the PRIMARY screen and C0 recovery only as fallback. Found by rendering PDF figures and reading them visually - a capability previously and wrongly described as "manual figure reading" and deferred to Mulham.

high F17 — Zhang 2021 rows are extrapolations with an unphysical intercept [resolved]

EVIDENCE

Figure 8 read visually. The slope reproduces the published Qmax exactly ($1/11.196 = 89.32$; $1/22.464 = 44.52$), but both intercepts are NEGATIVE (-0.5365, -2.0942), which in $C_e/q_e = 1/(Q_{\max} \cdot b) + C_e/Q_{\max}$ implies $b < 0$ and is unphysical; the table nonetheless reports positive b. Data reach only $C_e = 3.45$ and 3.75 mg/L, giving $\theta = 0.365$ and 0.074 - the slag fit reaches 7% of its plateau, worse than the chabazite artefact at 17%.

ACTION

Both rows flagged as extrapolations and excluded from modelling alongside chabazite. Screen now COMPLETE at 35 of 35: 31 sound, 1 borderline, 3 extrapolated.

high F18 — The Oulu 29Si deconvolution was already published; it explains our failed fit [resolved]

EVIDENCE

Table 2 of Hossain 2026 (doi:10.1016/j.matdes.2026.115471, page 6) reports the deconvolved Si environments for all 12 selected samples - the values our own fit attempted to derive. Found by rendering the page and reading it. The authors use TEN environments (Q0, Q1, Q2(1Al), Q2(0Al), Q4(4Al), Q3(1Al), Q4(3Al), Q3/Q4(2Al), Q3/Q4(1Al), Q4(0Al)) including Q3 and Q2 species with non-bridging oxygens. Our model used FIVE Q4(mAl) Gaussians only, which cannot represent that intensity; the optimiser pushed it outward and drove Q4(3Al) to zero.

ACTION

Confirms the NMR rejection was correct AND for the right reason - the chemical sanity check detected a structurally wrong peak model that $R^2 = 0.998$ and both pre-registered gates could not. Pipeline closed. The Oulu letter is no longer needed for this purpose. Values transcribed into src/geomind/data/nmr_ari.py; 11 of 12 rows sum to exactly 100%, one to 99.0 (published as such).

high F19 — The Al-richness descriptor is valid only within a structural class, and the Oulu backfill does not widen its evidence base [resolved]

EVIDENCE

Backfilling ARI from F18 attached the descriptor to 39 pool rows / 12 sorbents. Split by structure, the 12 sorbents behave as two different materials. Ca-FREE (n=4, Q4 45-73%): ARI +0.932, framework-only ARI +0.940 - uptake spans 2.13-11.53 mg/g (sd 3.98). Ca-BEARING (n=8, Q4 0-26%, Q0-Q2 dominated): ARI -0.108, Si/Al -0.135, Ca/Al -0.104, Q4% -0.247, BET +0.462 - NOTHING predicts, and uptake is nearly flat at 7.27 +/- 1.31 mg/g regardless of composition. Pooling all 12 gives ARI +0.536, a mediocre value that conceals two regimes. Ca acts as a network modifier: there is no condensed aluminosilicate framework for a framework descriptor to measure, and uptake proceeds by a different mechanism (C-A-S-H exchange), roughly composition-independent.

ACTION

Restrict the ARI claim to Q4-rich framework aluminosilicates and state the structural gate explicitly. Generalises F9 one level down: never pool ADSORBATE classes, and never pool SORBENT structural classes either. IMPORTANT correction to my own estimate - I projected this backfill would widen the descriptor to ~19 sorbents; in fact only 4 of the 12 lie in the regime where it applies, so the evidence base is unchanged at Varon n=7 (Sr) + Oulu n=4 (NH4+). Al^IV scarcity remains the binding constraint on M4/M5.

critical

F20 — NMR characterisation and adsorption-capacity measurement are near-disjoint in this literature - Al^{IV} scarcity is structural, not a sampling accident

[open]

EVIDENCE

Corpus-wide classification of all 30 PDFs by keyword balance (adsorption terms vs leaching/immobilisation terms) against NMR content. Of the 6 papers carrying substantial solid-state NMR, FOUR are immobilisation/wasteform studies (Nevin 2026 nmr=54, Kim 2026 nmr=66, Vettese 2025 nmr=33, Kurumisawa 2021 nmr=13), ONE is adsorption (Hossain/Oulu nmr=11, but NH4+ and dyes only - no Cs/Sr), and ONE is mixed (Varon 2025 nmr=32). Meanwhile ALL 13 papers classified as adsorption studies report ZERO solid-state NMR. Varon is the single bridge between the two - which is exactly why Al^{IV} occurs in precisely 7 pool rows, all from Varon. These are two different research communities: water-treatment groups measure isotherms and BET and never run MAS NMR; nuclear-wasteform groups run full 29Si/27Al/133Cs NMR and measure leach rates and diffusivities, never isotherms.

ACTION

More scraping CANNOT fix Al^{IV} scarcity - the co-occurrence is rare by the structure of the field, not by chance, so Q7 has no answer within the adsorption literature. Two routes remain: (a) measure NMR ourselves on sorbents of known capacity, or (b) REFRAME the M4/M5 target property from adsorption capacity to leach resistance, where NMR is abundant. Note that route (b) reverses earlier triage - Kurumisawa 2021 and J. Environ. Manage. 370 (2024) were both REJECTED as "diffusivity/leaching studies", i.e. we discarded precisely the papers richest in the descriptor we need. The project aim ("immobilisation of Cs-137/Sr-90") is arguably better served by leach resistance than by adsorption capacity anyway; the two have been conflated. Decision required from PI - see reports/q7-descriptor-hunt-report.md.

high

F21 — Sr_Leaching_Data audited against source for the first time - LI verified, De corrupted to zero, and the Yildirim rows are misattributed

[resolved]

EVIDENCE

Both cited sources (Kim 2026, Nevin 2026) are now known to be on disk (see F13), so this sheet became auditable. Line-by-line against Kim Table 2 (p10) and Nevin p6: (a) LEACHABILITY INDEX - VERIFIED, 6 of 6 determinable rows exact. Kim NaSr1 15.5, NaSr2 14.8, NaSr3 16.0, KSr3 16.0; Nevin Sr_1 18.7, Sr_3 19.1. KSr1/KSr2 are correctly recorded as below detection - Kim states Sr leachate was under the ICP-OES limit (0.001 ppm), yielding non-detectable De and LI. (b) De_cm2_s - CORRUPTED. Every row reads 0.0, whereas Kim Table 2 reports 8.88E-16, 3.80E-14, 2.10E-16, 6.36E-16, 5.80E-18 and so on. Precision was lost to zero. Kim's De is also TIME-RESOLVED over five points (2 h, 7 h, 7 d, 14 d, 21 d), so no single scalar De pairs with the averaged LI in the first place. (c) CN_Sr_EXAFS on the two Yildirim_2024 rows - MISATTRIBUTED. The sheet gives Sr_Pre_1 = 7.87 and Sr_Post_1 = 7.95 sourced to Yildirim_2024. Kim p26 states the NaSr3 coordination number "was 7.87 +/- 0.69 before the leaching test" and "7.95 +/- 0.43 afterwards" - these are Kim's own NaSr3 pre/post values, and Kim attributes the pre-leach figure to reference [7], which is Geddes et al., J. Hazard. Mater. 488 (2025) 137426 - NOT Yildirim. The KSr3 value 7.55 is correctly attributed.

ACTION

Promote the LI column to VERIFIED_TRUE - it is the first immobilisation target variable in the project to survive source audit. Quarantine De_cm2_s entirely (re-extract from Kim Table 2 as a time series if needed). Quarantine both Yildirim_2024 rows as misattributed - third instance of the F7 pattern.

high

F22 — Pool B built and audited - LI verified twice over, but it has almost no compositional variation paired with NMR

[resolved]

EVIDENCE

Pool B v1 = 16 rows / 2 sources (Kim 2026, Nevin 2026), of which 8 carry a retention target, 2 are left-censored, and 6 are structure-only. (a) The LI = -log10(De) relation is now confirmed against TWO independent published tables - Kim Table 2 (12 of 13 cells to within 0.06) and Nevin Table 2 (4 of 4: 18.66/19.11/13.27/13.22 vs published 18.7/19.1/13.3/13.2). (b) A PUBLISHED ERRATUM was found. Kim Table 2 cell NaSr1 @ 2 h prints "8.88 E-16 (15.5)" but -log10(8.88e-16) = 15.05. Kim's own column Average LI of 15.5 equals mean(15.05, 15.94) exactly; had the cell truly been 15.5 the average would read 15.7. The printed (15.5) is a typo for (15.1). Verified at 430 dpi, so it is not a rendering artefact. The column average is unaffected. (c) The master sheet Sr_Leaching_Data carried only Nevin's two Sr rows and SILENTLY DROPPED both Cs rows (Cs_1 LI 13.3, Cs_3 LI 13.2) - a 50% loss of that table, undetectable without reading the source. (d) Nevin's headline result is preserved: in the SAME K-A-S-H gel, Sr is retained roughly six orders of magnitude better than Cs (De 7.8e-20 vs 6.0e-14). Nuclide identity dominates structure entirely at fixed composition.

ACTION

LIMITATION, stated plainly: only 4 rows pair an NMR descriptor with a retention target, and across those 4 the ARI is nearly constant (2.738-2.855) because Nevin is ONE formulation at Si/Al = 3. Kim varies activator (Na vs K) and loading but reports no Q4 deconvolution. So Pool B is currently as descriptor-starved as Pool A, for the opposite reason - Pool A has composition without NMR, Pool B has NMR without composition. Neither supports a descriptor model yet. This raises the priority of Q8 (Geddes 2025) further: it is the only identified source that may carry composition, NMR and a retention/binding measure together.

medium

F23 — Not every published deconvolution assigns m uniquely - Vettese's ARI is indeterminate and must not be computed

[resolved]

EVIDENCE

Vettese 2025 Table 4 (p7) reports a 29Si deconvolution, but its first two columns carry AMBIGUOUS assignments: "Q3(1Al) or Q4(4Al)" (m = 1 or 4) and "Q3 or Q4(3Al)" (m = 0 or 3). Computing the Al-richness index therefore has two permitted answers for the unleached sample: ARI = 2.289 if the Q4 readings are taken, or 0.861 if the Q3 readings are - a factor of 2.66. Populations do sum correctly (100.0 / 100.1 / 100.1), so the table is internally sound; it is the ASSIGNMENT that is undetermined, not the fit. Hossain and Nevin both assign m uniquely; Vettese does not.

ACTION

Do NOT compute or store an ARI for Vettese. A pipeline that derived ARI from any Q^n(mAl) table without checking assignment uniqueness would silently inject a 2.7x error into the one descriptor the project depends on. Vettese can still contribute structure-only rows (raw populations) and its Cs leach data, which is figure-only and single-formulation. Any future adapter must carry an assignment-ambiguity flag.

critical

F24 — Kurumisawa 2021 recovered - a rejected paper that bridges both pools - and it exposes a 12-order-of-magnitude naming collision in "diffusion coefficient" [resolved]

EVIDENCE

Kurumisawa 2021 (doi:10.1617/s11527-021-01758-y) was REJECTED during batch triage as "a diffusivity study (diffus 65)". Reading it inline shows that verdict was wrong on content: (a) Figure 3b publishes the Langmuir MONOLAYER Cs CAPACITY for five metakaolin geopolymers made with five different alkali activators (K11 0.315, K9 0.425, 0.66K 0.530, N11 0.727, KN11 0.393 mmol/g) - a Pool A capacity WITH real compositional variation, which Pool A is short of. Pool A: 121 -> 126 rows. (b) Figure 2 publishes a Cs diffusivity for the same five specimens - Pool B. Pool B: 16 -> 26 rows. (c) THE COLLISION. Kurumisawa's is an INGRESS coefficient (Cs diffusing INTO an immersed specimen), reported in m2/s and of order 4e-12 m2/s = 4e-8 cm2/s. Kim's and Nevin's are ANSI/ANS 16.1 EFFECTIVE diffusivities (nuclide leaching OUT of a doped wasteform), of order 1e-16 and 1e-19 cm2/s. Same name, opposite experiment, TWELVE orders of magnitude apart. (d) Table 3 is the 29Si NMR of the ACTIVATOR SOLUTION (Q0-Q3 of the silicate), NOT of the hardened gel - so this paper supplies no framework Al^{IV} despite being NMR-bearing. (e) Applying the F14/F16 saturation screen to the recovered capacities: K11 theta 0.962 sound, KN11 0.847 sound, K9 0.704 borderline, 0.66K 0.608 borderline, and N11 theta 0.499 an EXTRAPOLATION ARTEFACT - N11 being the HIGHEST reported capacity of the five.

ACTION

Both halves ingested through separate adapters under D13. RetentionType gains INGRESS_DIFFUSIVITY as a distinct member and pooling_warning() now raises an explicit CATEGORY ERROR if both diffusivity kinds appear together. N11 flagged as an artefact. All Kurumisawa rows carry from_figure=True. LESSON: the batch triage rejected this paper on keyword balance without reading it, and keyword balance cannot see a capacity that lives only in a figure. The other triage rejections should be re-read the same way - starting with J. Environ. Manage. 370 (2024).

medium

F25 — The last triage rejection (J. Environ. Manage. 370) is upheld as data, but it is a high-value BIBLIOGRAPHY that names the paired-data sources both pools lack [resolved]

EVIDENCE

D15 re-read of Mukiza et al., J. Environ. Manage. 370 (2024) 122746, "Recent advances in immobilization of Cs/Sr-bearing wastes in AAMs: a review". It is a REVIEW, not primary data: zero Langmuir/Qmax/Kd of its own, and its Table 2 explicitly states "the data are not directly comparable". So the batch rejection was CORRECT for direct ingestion - none of its tabulated numbers may enter either pool as primary rows. BUT reading it surfaced exactly the paired-descriptor sources F20/F22 said were rare: (a) Komljenovic et al. 2020, J. Hazard. Mater. 388, doi:10.1016/j.jhazmat.2019.121765, "Immobilization of cesium with alkali-activated blast furnace slag" - the review states it ties 29Si NMR speciation (Q1, Q2(0Al), Q2(1Al) sites, Al^{IV}/Si of C-A-S-H) to strength AND Cs. NMR + composition + Cs in one paper. Same group (Provis) as Nevin/Geddes. A second candidate F20 bridge alongside Geddes 2025. (b) Qian et al. 2001, J. Nucl. Mater. 299, 199-204, doi:10.1016/S0022-3115(01)00700-0 - Cs and Sr distribution ratios R_d (3.9e3->6.4e3 mol/g Cs; 6.4e3->1.2e4 mol/g Sr). A K_d measure = Pool A. (c) Jang et al. 2016, J. Hazard. Mater. 318, 339-346, doi:10.1016/j.jhazmat.2016.07.003, "Physical barrier effect ... on diffusivity of Cs and Sr" - a Cs/Sr diffusivity (Pool B) source, and it is ALREADY ON DISK (papers/references/Physical barrier effect ...pdf), registered but never mined.

ACTION

Keep the review out of both pools as a data source. Use it as a source index. Q9 opened for Komljenovic 2020 (acquire), Q10 for Qian 2001 (acquire), and Jang 2016 queued for Pool B extraction since it is already held. This is the third time (after F24 Kurumisawa and the Kim/Nevin content-index miss) that real value was found by reading a document already in hand rather than acquiring a new one.

medium

F26 — Jang 2016 mined into Pool B - 14 rows, both nuclides, with pore structure; and it confirms the LI aggregation trap a third time [resolved]

EVIDENCE

Jang, Park & Lee, J. Hazard. Mater. 318 (2016) 339-346, doi:10.1016/j.jhazmat.2016.07.003 - the Q11 source, already on disk, now extracted. Seven doped waste forms (F1/F2/F3 fly-ash geopolymers, S1/S3/S5 fly-ash+slag blends, PC Portland baseline), each with Cs AND Sr leachability indices (Table 7) and pore structure (Table 5). Pool B: 26 -> 40 rows / 4 sources. (a) LI AGGREGATION CONFIRMED a third time. Jang's LI = mean over intervals of -log10(De_n): F1 Cs interval De's give 10.02 ~ published 10.0, whereas -log10(mean De) = 9.46 would be wrong by 0.5. Vindicates schema trap 4. (b) Sr LI > Cs LI in EVERY one of the 7 specimens - the universal Cs<Sr retention result, now across 7 more matrices, consistent with Nevin's ~6-order gap. (c) Si/Al deliberately NOT stored. The precursor-bulk Si/Al is computable (1.55-2.13) but omits the sodium-silicate activator Si, so it is not the gel Si/Al, and the F/S series use different silicate loadings (0.25 vs 0.17) so it is not even comparable across specimens. Storing it would repeat the F1 transplant error in a new form. Recorded as provenance-only constants. (d) Three new pore-structure columns added (porosity, capillary pore volume, critical pore diameter); Jang links these to De. corr(porosity, Cs LI) = -0.27 across the 6 geopolymers - weak, consistent with Jang's own point that critical pore DIAMETER matters more than total porosity.

ACTION

Ingested via _from_jang2016 as 14 LI rows. Mean De values retained as documented adapter constants (uncorrupted, unlike Kim's) but not emitted as separate rows, to avoid implying an independent measurement - a diffusivity target can regenerate them later. PC baseline flagged matrix_class=portland_cement, trivially filterable. Q11 closed. Fourth on-disk recovery of the session.

Appendix B

Pool A — every adsorption row

All 126 rows across 11 sources. ‘—’ = not reported for that row. Generated directly from the pipeline; nothing hand-entered.

#	source	sorbent	adsorbate	class	cap (mg/g)	type	Si/Al	BET	Al ^{IV} %	provenance
1	literature_compilation	Na-MKBFS	Cs	radionuclide	75.8	langmuir_qmax	2.13	—	—	El-Naggar_2018
2	literature_compilation	Na-MKBFS	Cs	radionuclide	80.5	langmuir_qmax	2.13	—	—	El-Naggar_2018
3	literature_compilation	Na-MKBFS	Cs	radionuclide	82.2	langmuir_qmax	2.13	—	—	El-Naggar_2018
4	literature_compilation	K-MKBFS	Cs	radionuclide	42.2	langmuir_qmax	2.4	—	—	El-Naggar_2018
5	literature_compilation	K-MKBFS	Cs	radionuclide	48	langmuir_qmax	2.4	—	—	El-Naggar_2018
6	literature_compilation	K-MKBFS	Cs	radionuclide	61.5	langmuir_qmax	2.4	—	—	El-Naggar_2018
7	literature_compilation	MU-PMK80	Cs	radionuclide	59.6	langmuir_qmax	1.43	—	—	Hamed_2025
8	literature_compilation	MU-PMK80	Eu	other	18.6	langmuir_qmax	1.43	—	—	Hamed_2025
9	literature_compilation	MU-PMK80	Cs	radionuclide	59.4	langmuir_qmax	1.43	—	—	Hamed_2025
10	literature_compilation	MU-PMK80	Eu	other	20	langmuir_qmax	1.43	—	—	Hamed_2025

#	source	sorbent	adsorbate	class	cap (mg/g)	type	Si/Al	BET	Al ^{IV} %	provenance
11	literature_compilation	MU-PMK80	Cs	radionuclide	57.7	langmuir_qmax	1.43	—	—	Hamed_2025
12	literature_compilation	MU-PMK80	Eu	other	21.6	langmuir_qmax	1.43	—	—	Hamed_2025
13	literature_compilation	MU-PMK80	Cs	radionuclide	59.6	langmuir_qmax	1.43	—	—	Hamed_2025
14	literature_compilation	MU-PMK80	Cs	radionuclide	59.4	langmuir_qmax	1.43	—	—	Hamed_2025
15	literature_compilation	MU-PMK80	Cs	radionuclide	57.7	langmuir_qmax	1.43	—	—	Hamed_2025
16	literature_compilation	MU-PMK80	Cs	radionuclide	59.6	langmuir_qmax	1.43	—	—	Hamed_2025
17	literature_compilation	FA-Geopolymer	Cs	radionuclide	112	langmuir_qmax	2	—	—	Tian_2019
18	literature_compilation	FA-Geopolymer	Sr	radionuclide	24.2	langmuir_qmax	2	—	—	Tian_2019
19	literature_compilation	Zeolite-Geopolymer	Cs	radionuclide	87.9	langmuir_qmax	2	—	—	Zheng_2023
20	literature_compilation	M/SZMs	Cs	radionuclide	104	freundlich_qmax	2	—	—	Lei_2021
21	literature_compilation	M/SZMs	Sr	radionuclide	54.9	langmuir_qmax	2	—	—	Lei_2021
22	oulu2026	Si20 Al1 Na1	NH4+	analogue	2.13	single_point_q	20	249	—	doi:10.1016/j.matde...
23	oulu2026	Si15 Al1 Na1	NH4+	analogue	4.27	single_point_q	15	242	—	doi:10.1016/j.matde...
24	oulu2026	Si10 Al1 Na1	NH4+	analogue	5.47	single_point_q	10	206	—	doi:10.1016/j.matde...
25	oulu2026	Si5Al1Na1	NH4+	analogue	7	single_point_q	5	132	—	doi:10.1016/j.matde...
26	oulu2026	Si4Al1Na1	NH4+	analogue	8.07	single_point_q	4	109	—	doi:10.1016/j.matde...
27	oulu2026	Si3Al1Na1	NH4+	analogue	9.13	single_point_q	3	71.6	—	doi:10.1016/j.matde...
28	oulu2026	Si2Al1Na1	NH4+	analogue	11.5	single_point_q	2	31	—	doi:10.1016/j.matde...
29	oulu2026	Si1Al1Na1	NH4+	analogue	13.7	single_point_q	1	8.27	—	doi:10.1016/j.matde...
30	oulu2026	Si20 Al1 Na1 Ca21	NH4+	analogue	7.73	single_point_q	20	6.37	—	doi:10.1016/j.matde...
31	oulu2026	Si15 Al1 Na1 Ca16	NH4+	analogue	6.87	single_point_q	15	6.04	—	doi:10.1016/j.matde...
32	oulu2026	Si10 Al1 Na1 Ca11	NH4+	analogue	5.33	single_point_q	10	5.14	—	doi:10.1016/j.matde...
33	oulu2026	Si5 Al1 Na1 Ca6	NH4+	analogue	4.53	single_point_q	5	3.52	—	doi:10.1016/j.matde...
34	oulu2026	Si4 Al1 Na1 Ca5	NH4+	analogue	5.93	single_point_q	4	6.92	—	doi:10.1016/j.matde...
35	oulu2026	Si3 Al1 Na1 Ca4	NH4+	analogue	6.67	single_point_q	3	10.3	—	doi:10.1016/j.matde...
36	oulu2026	Si2 Al1 Na1 Ca3	NH4+	analogue	8.87	single_point_q	2	13.2	—	doi:10.1016/j.matde...
37	oulu2026	Si1 Al1 Na1 Ca2	NH4+	analogue	11	single_point_q	1	14.1	—	doi:10.1016/j.matde...
38	oulu2026	Si4 Al1 Na1 Ca1	NH4+	analogue	6.53	single_point_q	4	2.79	—	doi:10.1016/j.matde...
39	oulu2026	Si4 Al1 Na1 Ca2	NH4+	analogue	6.8	single_point_q	4	1.03	—	doi:10.1016/j.matde...
40	oulu2026	Si4 Al1 Na1 Ca3	NH4+	analogue	8.27	single_point_q	4	2.7	—	doi:10.1016/j.matde...
41	oulu2026	Si4 Al1 Na1 Ca4	NH4+	analogue	8.67	single_point_q	4	7.95	—	doi:10.1016/j.matde...
42	oulu2026	Si4 Al1 Na1 Ca5	NH4+	analogue	5.93	single_point_q	4	6.92	—	doi:10.1016/j.matde...
43	oulu2026	Si20 Al1 Na1	methylene_blue	dye	90.3	single_point_q	20	249	—	doi:10.1016/j.matde...
44	oulu2026	Si15 Al1 Na1	methylene_blue	dye	87.6	single_point_q	15	242	—	doi:10.1016/j.matde...
45	oulu2026	Si10 Al1 Na1	methylene_blue	dye	77.2	single_point_q	10	206	—	doi:10.1016/j.matde...
46	oulu2026	Si5Al1Na1	methylene_blue	dye	56.6	single_point_q	5	132	—	doi:10.1016/j.matde...
47	oulu2026	Si4Al1Na1	methylene_blue	dye	48.4	single_point_q	4	109	—	doi:10.1016/j.matde...
48	oulu2026	Si3Al1Na1	methylene_blue	dye	36.7	single_point_q	3	71.6	—	doi:10.1016/j.matde...
49	oulu2026	Si2Al1Na1	methylene_blue	dye	14.3	single_point_q	2	31	—	doi:10.1016/j.matde...
50	oulu2026	Si1Al1Na1	methylene_blue	dye	3.57	single_point_q	1	8.27	—	doi:10.1016/j.matde...
51	oulu2026	Si20 Al1 Na1 Ca21	methylene_blue	dye	27	single_point_q	20	6.37	—	doi:10.1016/j.matde...
52	oulu2026	Si15 Al1 Na1 Ca16	methylene_blue	dye	25.2	single_point_q	15	6.04	—	doi:10.1016/j.matde...
53	oulu2026	Si10 Al1 Na1 Ca11	methylene_blue	dye	21.6	single_point_q	10	5.14	—	doi:10.1016/j.matde...
54	oulu2026	Si5 Al1 Na1 Ca6	methylene_blue	dye	14.2	single_point_q	5	3.52	—	doi:10.1016/j.matde...
55	oulu2026	Si4 Al1 Na1 Ca5	methylene_blue	dye	20.3	single_point_q	4	6.92	—	doi:10.1016/j.matde...
56	oulu2026	Si3 Al1 Na1 Ca4	methylene_blue	dye	26.8	single_point_q	3	10.3	—	doi:10.1016/j.matde...
57	oulu2026	Si2 Al1 Na1 Ca3	methylene_blue	dye	31.4	single_point_q	2	13.2	—	doi:10.1016/j.matde...
58	oulu2026	Si1 Al1 Na1 Ca2	methylene_blue	dye	34.9	single_point_q	1	14.1	—	doi:10.1016/j.matde...
59	oulu2026	Si4 Al1 Na1 Ca1	methylene_blue	dye	11.2	single_point_q	4	2.79	—	doi:10.1016/j.matde...

#	source	sorbent	adsorbate	class	cap (mg/g)	type	Si/Al	BET	Al ^{IV} %	provenance
60	oulu2026	Si4 Al1 Na1 Ca2	methylene_blue	dye	9.3	single_point_q	4	1.03	—	doi:10.1016/j.matde...
61	oulu2026	Si4 Al1 Na1 Ca3	methylene_blue	dye	11.5	single_point_q	4	2.7	—	doi:10.1016/j.matde...
62	oulu2026	Si4 Al1 Na1 Ca4	methylene_blue	dye	27.2	single_point_q	4	7.95	—	doi:10.1016/j.matde...
63	oulu2026	Si4 Al1 Na1 Ca5	methylene_blue	dye	20.3	single_point_q	4	6.92	—	doi:10.1016/j.matde...
64	oulu2026	Si20 Al1 Na1	rhodamine_6G	dye	53.7	single_point_q	20	249	—	doi:10.1016/j.matde...
65	oulu2026	Si15 Al1 Na1	rhodamine_6G	dye	28	single_point_q	15	242	—	doi:10.1016/j.matde...
66	oulu2026	Si10 Al1 Na1	rhodamine_6G	dye	13.5	single_point_q	10	206	—	doi:10.1016/j.matde...
67	oulu2026	Si5Al1Na1	rhodamine_6G	dye	7.74	single_point_q	5	132	—	doi:10.1016/j.matde...
68	oulu2026	Si4Al1Na1	rhodamine_6G	dye	6.29	single_point_q	4	109	—	doi:10.1016/j.matde...
69	oulu2026	Si3Al1Na1	rhodamine_6G	dye	5.93	single_point_q	3	71.6	—	doi:10.1016/j.matde...
70	oulu2026	Si2Al1Na1	rhodamine_6G	dye	5.21	single_point_q	2	31	—	doi:10.1016/j.matde...
71	oulu2026	Si1Al1Na1	rhodamine_6G	dye	2.31	single_point_q	1	8.27	—	doi:10.1016/j.matde...
72	oulu2026	Si20 Al1 Na1 Ca21	rhodamine_6G	dye	36.3	single_point_q	20	6.37	—	doi:10.1016/j.matde...
73	oulu2026	Si15 Al1 Na1 Ca16	rhodamine_6G	dye	33.4	single_point_q	15	6.04	—	doi:10.1016/j.matde...
74	oulu2026	Si10 Al1 Na1 Ca11	rhodamine_6G	dye	27.6	single_point_q	10	5.14	—	doi:10.1016/j.matde...
75	oulu2026	Si5 Al1 Na1 Ca6	rhodamine_6G	dye	12.4	single_point_q	5	3.52	—	doi:10.1016/j.matde...
76	oulu2026	Si4 Al1 Na1 Ca5	rhodamine_6G	dye	14.6	single_point_q	4	6.92	—	doi:10.1016/j.matde...
77	oulu2026	Si3 Al1 Na1 Ca4	rhodamine_6G	dye	17.1	single_point_q	3	10.3	—	doi:10.1016/j.matde...
78	oulu2026	Si2 Al1 Na1 Ca3	rhodamine_6G	dye	18.9	single_point_q	2	13.2	—	doi:10.1016/j.matde...
79	oulu2026	Si1 Al1 Na1 Ca2	rhodamine_6G	dye	20.4	single_point_q	1	14.1	—	doi:10.1016/j.matde...
80	oulu2026	Si4 Al1 Na1 Ca1	rhodamine_6G	dye	9.18	single_point_q	4	2.79	—	doi:10.1016/j.matde...
81	oulu2026	Si4 Al1 Na1 Ca2	rhodamine_6G	dye	7.74	single_point_q	4	1.03	—	doi:10.1016/j.matde...
82	oulu2026	Si4 Al1 Na1 Ca3	rhodamine_6G	dye	10.3	single_point_q	4	2.7	—	doi:10.1016/j.matde...
83	oulu2026	Si4 Al1 Na1 Ca4	rhodamine_6G	dye	19.3	single_point_q	4	7.95	—	doi:10.1016/j.matde...
84	oulu2026	Si4 Al1 Na1 Ca5	rhodamine_6G	dye	14.6	single_point_q	4	6.92	—	doi:10.1016/j.matde...
85	katada2024	FAU 6.1	Cs	radionuclide	73.1	iec_meq_g	1.37	—	—	10.1021/acs.langmu...
86	katada2024	LTA 6.7	Cs	radionuclide	23.9	iec_meq_g	1.12	—	—	10.1021/acs.langmu...
87	katada2024	MFI 1.2	Cs	radionuclide	133	iec_meq_g	12.3	—	—	10.1021/acs.langmu...
88	katada2024	YFI 1.3	Cs	radionuclide	110	iec_meq_g	11.1	—	—	10.1021/acs.langmu...
89	katada2024	MOR 1.5	Cs	radionuclide	191	iec_meq_g	9.88	—	—	10.1021/acs.langmu...
90	katada2024	MOR 1.7	Cs	radionuclide	211	iec_meq_g	8.37	—	—	10.1021/acs.langmu...
91	tarnovsky2024	Kaolin	Cs	radionuclide	13.3	langmuir_q_max	1.03	9	—	10.15407/hftp15.01...
92	tarnovsky2024	Kaolin	Sr	radionuclide	14.9	langmuir_q_max	1.03	9	—	10.15407/hftp15.01...
93	tarnovsky2024	GP 1	Cs	radionuclide	233	langmuir_q_max	1.95	26	—	10.15407/hftp15.01...
94	tarnovsky2024	GP 1	Sr	radionuclide	48.2	langmuir_q_max	1.95	26	—	10.15407/hftp15.01...
95	tarnovsky2024	GP 2	Cs	radionuclide	124	langmuir_q_max	1.59	88	—	10.15407/hftp15.01...
96	tarnovsky2024	GP 2	Sr	radionuclide	57	langmuir_q_max	1.59	88	—	10.15407/hftp15.01...
97	varon2025	K-0.5-16	Sr	radionuclide	—	kd_mL_g	1.14	72	96	10.1016/j.clema.202...
98	varon2025	K-1-16	Sr	radionuclide	—	kd_mL_g	1.55	163	93.8	10.1016/j.clema.202...
99	varon2025	K-1.2-16	Sr	radionuclide	—	kd_mL_g	1.67	188	94	10.1016/j.clema.202...
100	varon2025	K-1.5-16	Sr	radionuclide	—	kd_mL_g	1.79	212	91.4	10.1016/j.clema.202...
101	varon2025	K-1.2-8 (inferred)	Sr	radionuclide	—	kd_mL_g	1.83	15	88	10.1016/j.clema.202...
102	varon2025	K-1.2-12 (inferred)	Sr	radionuclide	—	kd_mL_g	1.63	182	92.4	10.1016/j.clema.202...
103	varon2025	Na-1.2-16 (inferred)	Sr	radionuclide	—	kd_mL_g	1.67	38	84.7	10.1016/j.clema.202...
104	lin2026	ACSR	Sr	radionuclide	222	iec_meq_g	—	—	—	10.3390/ceramics902...
105	lin2026	NCGO	Cs	radionuclide	54.5	iec_meq_g	—	39.2	—	10.3390/ceramics902...
106	lin2026	NiGO	Cs	radionuclide	46.5	iec_meq_g	—	38.8	—	10.3390/ceramics902...
107	lin2026	CoGO	Cs	radionuclide	45.2	iec_meq_g	—	29.3	—	10.3390/ceramics902...
108	lin2026	Ba-4A	Sr	radionuclide	167	iec_meq_g	—	—	—	10.3390/ceramics902...

#	source	sorbent	adsorbate	class	cap (mg/g)	type	Si/Al	BET	Al ^{IV} %	provenance
109	lin2026	CuFC/4A	Cs	radionuclide	86.4	iec_meq_g	—	5.25	—	10.3390/ceramics902...
110	tian2019_espr	Zeolite A (from fly ash)	Cs	radionuclide	282	langmuir_q_max	0.821	121	—	10.1007/s11356-019-...
111	tian2019_espr	Zeolite A (from fly ash)	Sr	radionuclide	169	langmuir_q_max	0.821	121	—	10.1007/s11356-019-...
112	xiang2021	GF300	Cs	radionuclide	162	single_point_q	1.02	53.8	—	10.1016/j.jece.2021...
113	xiang2021	GF400	Cs	radionuclide	164	single_point_q	1.02	49.4	—	10.1016/j.jece.2021...
114	xiang2021	GF500	Cs	radionuclide	186	single_point_q	1.02	45.8	—	10.1016/j.jece.2021...
115	xiang2021	GF600	Cs	radionuclide	192	single_point_q	1.02	37.8	—	10.1016/j.jece.2021...
116	xiang2021	GF600	Cs	radionuclide	216	langmuir_q_max	1.02	37.8	—	10.1016/j.jece.2021...
117	zhang2021	Flyash-geomaterial	Cs	radionuclide	89.3	langmuir_q_max	—	—	—	10.3390/app11188407
118	zhang2021	Slag-geomaterial	Cs	radionuclide	44.5	langmuir_q_max	—	—	—	10.3390/app11188407
119	baek2018	chabazite	Cs	radionuclide	1.25e+03	langmuir_q_max	1.5	—	—	10.1016/j.micromeso...
120	baek2018	stilbite	Cs	radionuclide	76.3	langmuir_q_max	3.88	—	—	10.1016/j.micromeso...
121	baek2018	heulandite	Cs	radionuclide	59.5	langmuir_q_max	3.48	—	—	10.1016/j.micromeso...
122	kurumisawa2021	K11	Cs	radionuclide	41.9	langmuir_q_max	—	—	—	10.1617/s11527-021-...
123	kurumisawa2021	K9	Cs	radionuclide	56.5	langmuir_q_max	—	—	—	10.1617/s11527-021-...
124	kurumisawa2021	0.66K	Cs	radionuclide	70.4	langmuir_q_max	—	—	—	10.1617/s11527-021-...
125	kurumisawa2021	N11	Cs	radionuclide	96.6	langmuir_q_max	—	—	—	10.1617/s11527-021-...
126	kurumisawa2021	KN11	Cs	radionuclide	52.2	langmuir_q_max	—	—	—	10.1617/s11527-021-...

Appendix C

Pool B — every immobilisation row

All 40 rows across 4 sources. ‘target’ distinguishes a leachability index (higher = better) from an ingress diffusivity; the two are never pooled. ‘cens.’ marks a left-censored (below-detection, best-performing) result.

#	source	matrix	class	nuclide	target	value	cens.	ARI	poros.%	provenance
1	kim2026	NaSr1	na_geopolymer	Sr	leachability index	15.5		—	—	10.1016/j.cemconres...
2	kim2026	NaSr2	na_geopolymer	Sr	leachability index	14.8		—	—	10.1016/j.cemconres...
3	kim2026	NaSr3	na_geopolymer	Sr	leachability index	16		—	—	10.1016/j.cemconres...
4	kim2026	KSr1	k_geopolymer	Sr	leachability index	—	left	—	—	10.1016/j.cemconres...
5	kim2026	KSr2	k_geopolymer	Sr	leachability index	—	left	—	—	10.1016/j.cemconres...
6	kim2026	KSr3	k_geopolymer	Sr	leachability index	16		—	—	10.1016/j.cemconres...
7	nevin2026	C_0	k_geopolymer	none	—	—		2.81	—	10.1039/d5ta90211h
8	nevin2026	Sr_1	k_geopolymer	Sr	leachability index	18.7		2.75	—	10.1039/d5ta90211h
9	nevin2026	Sr_3	k_geopolymer	Sr	leachability index	19.1		2.8	—	10.1039/d5ta90211h
10	nevin2026	Cs_1	k_geopolymer	Cs	leachability index	13.3		2.74	—	10.1039/d5ta90211h
11	nevin2026	Cs_3	k_geopolymer	Cs	leachability index	13.2		2.85	—	10.1039/d5ta90211h
12	nevin2026	C_0	k_geopolymer	none	—	—		2.81	—	10.1039/d5ta90211h
13	nevin2026	Sr_1	k_geopolymer	Sr	—	—		2.93	—	10.1039/d5ta90211h
14	nevin2026	Sr_3	k_geopolymer	Sr	—	—		2.81	—	10.1039/d5ta90211h
15	nevin2026	Cs_1	k_geopolymer	Cs	—	—		2.8	—	10.1039/d5ta90211h
16	nevin2026	Cs_3	k_geopolymer	Cs	—	—		2.76	—	10.1039/d5ta90211h
17	kurumisawa2021	K11	k_geopolymer	Cs	d ingress cm2 s	4.70e-08		—	—	10.1617/s11527-021-...
18	kurumisawa2021	K11	k_geopolymer	Cs	d ingress cm2 s	4.00e-08		—	—	10.1617/s11527-021-...
19	kurumisawa2021	K9	k_geopolymer	Cs	d ingress cm2 s	7.00e-09		—	—	10.1617/s11527-021-...
20	kurumisawa2021	K9	k_geopolymer	Cs	d ingress cm2 s	4.80e-09		—	—	10.1617/s11527-021-...
21	kurumisawa2021	0.66K	k_geopolymer	Cs	d ingress cm2 s	4.40e-08		—	—	10.1617/s11527-021-...
22	kurumisawa2021	0.66K	k_geopolymer	Cs	d ingress cm2 s	4.00e-08		—	—	10.1617/s11527-021-...
23	kurumisawa2021	N11	na_geopolymer	Cs	d ingress cm2 s	8.30e-09		—	—	10.1617/s11527-021-...
24	kurumisawa2021	N11	na_geopolymer	Cs	d ingress cm2 s	4.70e-09		—	—	10.1617/s11527-021-...
25	kurumisawa2021	KN11	k_geopolymer	Cs	d ingress cm2 s	1.90e-08		—	—	10.1617/s11527-021-...
26	kurumisawa2021	KN11	k_geopolymer	Cs	d ingress cm2 s	2.00e-08		—	—	10.1617/s11527-021-...

#	source	matrix	class	nuclide	target	value	cens.	ARI	poros.%	provenance
27	jang2016	FA1 geopolymer	fly_ash_geopolymer	Cs	leachability index	10		—	29.9	10.1016/j.jhazmat.2...
28	jang2016	FA1 geopolymer	fly_ash_geopolymer	Sr	leachability index	14.1		—	29.9	10.1016/j.jhazmat.2...
29	jang2016	FA2 geopolymer	fly_ash_geopolymer	Cs	leachability index	10.1		—	33.9	10.1016/j.jhazmat.2...
30	jang2016	FA2 geopolymer	fly_ash_geopolymer	Sr	leachability index	14.1		—	33.9	10.1016/j.jhazmat.2...
31	jang2016	FA3 geopolymer	fly_ash_geopolymer	Cs	leachability index	10.2		—	31.9	10.1016/j.jhazmat.2...
32	jang2016	FA3 geopolymer	fly_ash_geopolymer	Sr	leachability index	14.4		—	31.9	10.1016/j.jhazmat.2...
33	jang2016	FA3+BFS 90:10	slag_blend	Cs	leachability index	7.1		—	39.7	10.1016/j.jhazmat.2...
34	jang2016	FA3+BFS 90:10	slag_blend	Sr	leachability index	11.5		—	39.7	10.1016/j.jhazmat.2...
35	jang2016	FA3+BFS 70:30	slag_blend	Cs	leachability index	8		—	31.9	10.1016/j.jhazmat.2...
36	jang2016	FA3+BFS 70:30	slag_blend	Sr	leachability index	12.7		—	31.9	10.1016/j.jhazmat.2...
37	jang2016	FA3+BFS 50:50	slag_blend	Cs	leachability index	8		—	25.9	10.1016/j.jhazmat.2...
38	jang2016	FA3+BFS 50:50	slag_blend	Sr	leachability index	12.1		—	25.9	10.1016/j.jhazmat.2...
39	jang2016	Portland cement	portland_cement	Cs	leachability index	7		—	28.3	10.1016/j.jhazmat.2...
40	jang2016	Portland cement	portland_cement	Sr	leachability index	9.8		—	28.3	10.1016/j.jhazmat.2...

Appendix D

Decisions register

The 16 standing project decisions, each with its reason. These are the rules the pipeline enforces.

#	Decision	Reason
D1	Drop Si_Al and Density from physical data	F1 - transplanted, not measured
D2	Re-scope M3b to cross-system transfer	Different material system; no viscosity; 92 mixes
D3	Repo private, all-rights-reserved	Unpublished research
D4	Copyrighted PDFs excluded from git	Copyright + 80 MB; manifest tracks them
D5	Do not use Feasibility_Ranges	F6 - anti-correlated with ground truth
D6	Kd kept in its own column, never in capacity_mg_g	Different physical quantity
D7	Reject Tian supplement's 24-row table	No composition; would pad without features
D8	Quarantine Na-MK, K-MK and all Niu 2022 rows	F7 - not in stated sources
D9	No subagents for extraction or analysis	Missing project context lets silent errors in
D10	Pre-register NMR acceptance at r>=0.9	Deconvolution is under-determined; stops retroactive tuning
D11	ARI claims are gated on Q4-rich framework sorbents	F19 - Ca-bearing gels have no framework; descriptor is meaningless there, not merely weak
D12	Never pool sorbent structural classes in a correlation	F19 - pooling hid ARI +0.932 and -0.108 behind a single +0.536
D13	Run adsorption capacity and leach resistance as TWO parallel pools, never merged	PI decision 2026-07-20 on F20. Removal-from-solution and immobilisation are different applications with different targets; the descriptor is available only in the second. Consistent with D12.
D15	Re-read every triage-rejected paper inline before trusting the rejection	F24 - Kurumisawa was rejected on keyword balance, but its Cs capacity lives only in a figure, which keyword counts cannot see
D16	Never compute ARI from a deconvolution whose m-assignment is ambiguous	F23 - Vettese's Table 4 admits ARI 2.289 or 0.861, a 2.7x fork
D14	Index the corpus by CONTENT, not filename	F13/F20 - Kim 2026 and Nevin 2026, the two most descriptor-rich papers held, were invisible to a filename-based check because their filenames are not in Latin script

Appendix E

Glossary of chemical terms

Term	Meaning
Alkali-activated material (AAM)	Binder made by activating an aluminosilicate precursor with a strong alkali; geopolymers are the aluminium-rich, low-calcium end of this family.
Al ^{IV} / framework aluminium	Aluminium in tetrahedral coordination within the network; each site carries a negative charge balanced by an exchangeable cation. The predictive descriptor of this project.
ARI (Al-richness index)	Mean number of Al neighbours per Si, $\sum_m m \cdot I_m / 100$, from ²⁹ Si NMR speciation; an NMR-based measure of framework aluminium.
BET surface area	Specific surface area from N ₂ physisorption (m ² g ⁻¹). Predicts dye (physisorption) uptake but <i>not</i> cation (ion-exchange) uptake.

Term	Meaning
Distribution coefficient K_d	Ratio of sorbed to solution concentration (mL g^{-1}); a measure of affinity, distinct from a capacity.
Effective diffusivity D_e	Rate constant for a nuclide leaching out of a waste form ($\text{cm}^2 \text{s}^{-1}$); lower is better.
Leachability index (LI, L)	$-\log_{10}(D_e)$; ANSI/ANS 16.1 ranking of immobilisation. Higher = better; ≥ 6 is the acceptance floor.
Langmuir Q_{max}	Fitted monolayer saturation capacity (mg g^{-1}); meaningful only if the experiment approached saturation (the θ screen).
$Q^n(\text{mAl})$	A silicon environment with n bridging oxygens, m of them to aluminium; from ^{29}Si MAS NMR.
R_L / θ	Langmuir separation factor; $\theta = 1 - R_L$ is the fraction of saturation reached, used to screen fitted capacities.
Si/Al ratio	Molar silicon-to-aluminium ratio; lower means more charge-balancing sites.
Network modifier (Ca)	Calcium disrupts the aluminosilicate framework toward low-connectivity C–A–S–H, where framework descriptors lose meaning.